

Research and domain understanding

U.S. purchase mortgages: customer clarity and complete handoffs define the opportunity

21%

of buyers were first-time buyers

NAR 2025 survey, July 2024–June 2025

>36%

rise in median total loan costs

Home purchase loans, 2021–2023. CFPB

3 business days

Loan Estimate delivery / mailing

Generally after receipt of all six TRID elements

People and needs

Loan officer + first-time buyers
Separate financial profiles and intentions.
Clear options, privacy and next steps.

Managers, processors, underwriters,
Compliance, pricing, IT/security and
authorized referral partners.

Journey and observed problem

Before: scattered history and offers.
During: discovery, estimates and tool switching.
After: re-entry, document requests and handoff.

Hypotheses to validate: inconsistent service,
stalled applications, unclear costs, repeated
requests and incomplete processing files.

Controls and open questions

Prevent unsupported eligibility claims,
steering, privacy failures and missed triggers.

Confirm joint intent, property and deadlines,
employment, lender/state authorization,
programs, consent, accessibility, retention
and ownership during system outages.

Scope assumption: an agent-led meeting. AI assists discovery and follow-through. Authorized lender processes make credit decisions.

The meeting journey and eight live nudges

Eight key examples of co-pilot’s effectiveness addressing the problem: gguidance stays private and facts remain unverified.

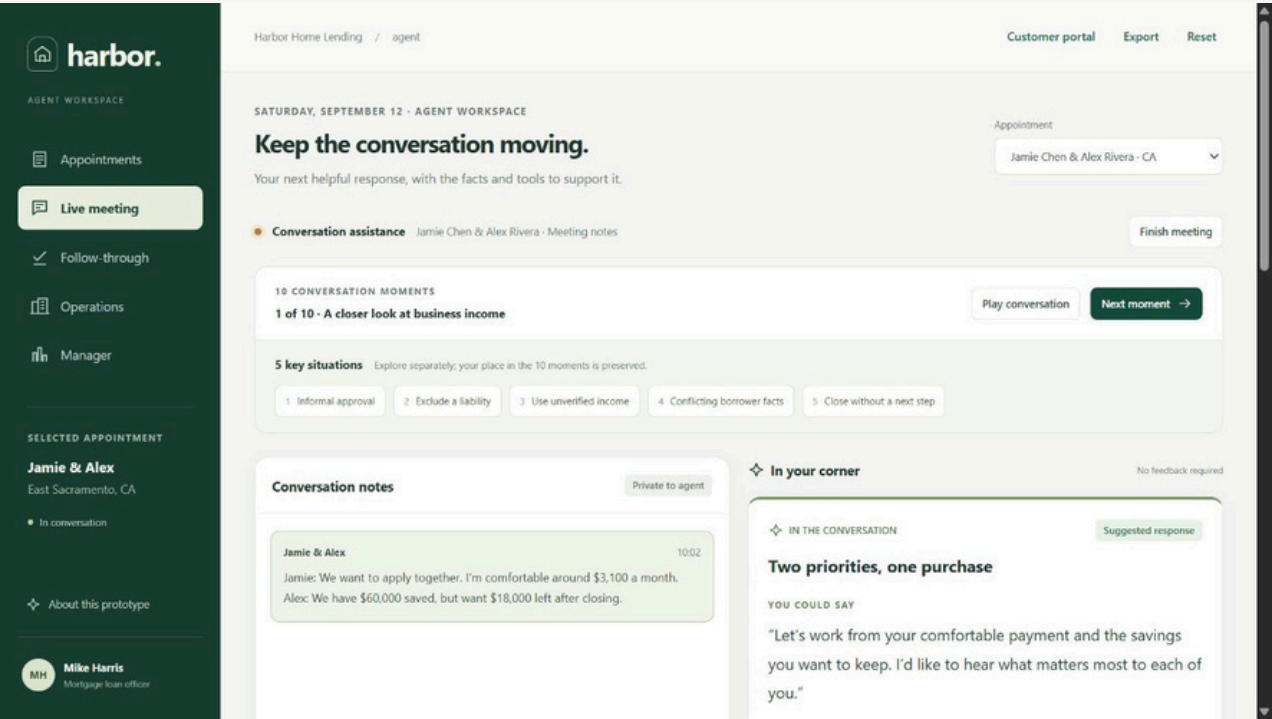
Stage	Manual process	Harbor copilot journey
Before	Scattered CRM notes, generic preparation	Profile-specific brief and agenda, known customer details handy
During	Conversation + notes + calculators + missed facts	Suggested sentence, supporting facts, relevant action, information capture
After / oversight	Re-entry, vague follow-up, late discovery of gaps	Reviewed recap, owned tasks, operations handoff, separate manager view

Moment / trigger	Nudge and agent action	Record / action generated	Risk if wrong
1 Payment vs savings	Ask each buyer about payment comfort and cash buffer	Separate stated priorities and joint intent	Misattributed preference
2 Different income paths	Explain specialist review before using business earnings	Income review task, conditional evidence request	Wrong document burden
3 “Which mortgage?”	Compare savings retained and payment trade-offs	Open example of financial illustrations	Narrow / stale comparison
4 Competitor rate	Compare points, fees, credits and lock details	Request written offer, review missing terms	Unfair cost comparison
5 Shared car debt	Confirm the same account before counting it once	Keep car + student payments for validation	Omitted / duplicated debt
6 Six elements received	Explain that the disclosure process can begin	Receipt event + owned task, independent of nudge	Missed disclosure clock
8 Cash after closing	Include closing costs and prepaids in the discussion	Reopen illustration and test savings remaining	False liquidity confidence
10 “Let’s follow up”	Agree purpose, owner, date, time and channel	Reviewed recap and follow-up plan	Unwanted contact / pressure

No mandatory live ratings. “Save for later” and optional specialist review preserve flow. Moments mark these live nudges in the prototype.

Prototype, integrations and action boundaries

Harbor is the Darwix AI prototype. Local workflows work today. External systems and live AI do not.



[Open Harbor prototype \(local server\)](#)

Two appointments: Jamie + Alex (CA), Maya relocating to Austin (TX).
Preparation, live meeting, recap, portal, operations and separate manager view.

Calculators and local state function.
Scripted AI, mocked writes and delivery.
No APR calculation. No file-content upload.
Role previews provide no authentication.

Receipt and deadline tracking must not wait for agent acceptance. Existing lender disclosure workflows remain authoritative.

Target system	Read	Write back	Today
CRM	History, contacts, bookings	Reviewed profile, outcome, tasks	Mock
LOS	Status, facts, requirements	Reviewed intake, gaps, handoff	Mock
Pricing / products	Rates, fees, program rules	Scenario request / reference	Fixed samples
Document portal	Requests, receipt, corrections	Approved recap, requests, updates	Names only
Calendar / messaging	Availability, permissions	Agreed meetings, reminders	No delivery
Operations	Owners, deadlines, task status	Assignments, review, escalations	Mock
Policy / identity / consent	Guidance, license, access, consent	Consent evidence, access log	Future
Speech / AI	Permitted talk + approved context	Suggestions, facts, summary drafts	Scripted
Analytics / archive	Events, outcomes, review history	Metrics, auditable records	Local only

AGENT APPROVAL

Publish recap, request documents, commit facts, submit handoff, confirm follow-up. Customer authority also applies to credit pulls / locks.

MAY AUTOMATE

Agenda, unverified capture, gap / conflict flags, drafts, internal alerts and already-authorized reminders.

NEVER AUTONOMOUS

Credit decisions, income verification, debt removal, conflict resolution, rate / date promises, invented consent or altered receipt evidence.

Three-week pilot and measurement

Proposed pilot: 3–5 agents, one branch/state, 30–50 meetings with individual and joint buyers, including variable income

30% less post-meeting administration

Including checking and corrections, without worse completeness or customer understanding. Proposed target, not a result.

Week 1	Week 2	Week 3
Train, establish baseline, test edge cases, validate permissions and data	Supervised meetings. Review every recap, material warning and system write post meeting. Limited in-meeting nudges	Continue controlled use. Assess results, interview customers / agents, decide expansion.

Pilot delivery boundary

Functional: build before pilot	Simulation / deferral
Capture each buyer’s details, flag conflicting information, let agents review summaries, update one CRM, and track follow-up tasks	For final loan-system submission lender’s existing systems still perform the real work.
Priorities: 1. accurate capture 2. improved response (better RAG, fine tuning) 3. improved handoff and pre-post tasks 4. integrations	Defer credit decisions, pulls / locks, recommendations, multi-state and wider integration.

Metric family	Proposed targets / measures
Product	≥80% eligible-meeting use. ≥90% next step or explicit decline. Customer clarity.
Business	30% admin-time reduction, repeat requests, seven-day application completion.
AI quality	≥98% material-field accuracy, ≥80% useful nudges, 100% critical test-set detection, latency.
Compliance	100% required consent / approval evidence. Zero unauthorized writes or material incidents. Matched-case consistency.

Entry gate: secure storage, authenticated access, consent and approved services/connectors ready.

Pause for material privacy breaches, unauthorized decisions/writes or missed critical safeguards. Do not expand if accuracy, adoption or time-saving targets fail.

Five independent situations and safe responses

Analysing five critical cases and design decisions for them. “Why this appears” explains the intervention and its risk.

Situation / trigger	Private guidance and agent response	Preserved evidence / next action	Risk if mishandled
Informal approval “Sounds like we’re approved”	Correct the expectation: review is incomplete. Agent clarifies the actual status without promising approval.	Record the correction and open questions. Authorized lender process owns any formal decision.	Buyer acts on an approval that does not exist.
Excluding debt “Can we leave that loan out?”	Keep the obligation visible. Agent explains that a specialist must review the applicable treatment.	Retain the debt. Create an owned review task. Never silently alter the inputs.	Overstated capacity or false reassurance.
Unverifiable income “Use what I say I earn”	Separate stated earnings from qualifying income. Agent explains the evidence and review needed.	Keep stated amount with source. Leave qualifying income unresolved. Request specialist review.	Unsupported affordability or unnecessary discouragement
Conflicting facts A later amount differs	Show both statements and ask a neutral clarification. Agent records the answer, source and reason.	Preserve both versions. Keep the existing value pending clarification. Re-review changed intake.	Invented / averaged facts distort the file.
Closing without a next step “Thanks, that’s all”	Ask for an action, owner and date, or respect an explicit no-follow-up choice.	Record purpose, time zone, channel and permission. No assumed meeting or forced reminder.	Stalled handoff or unwanted pressure.

The control pattern

Preserve source facts. Offer a correction or review action. Let the agent resolve material issues. Keep customer choice explicit.

Agent review is not income verification. A review task is not a compliance finding. Reg B notifications stay with authorized lender workflows.

Launch decision: Meeting Copilot Lite

Trustworthy capture, deterministic routing and accountable follow-through take priority over richer AI and dashboard features

Launch	Simplify	Defer	Reject
Separate buyer notes and configurable discovery checklist.	Optional transcription with manual notes.	Unavailable integrations, live speech coaching and generative recommendations.	Raw audio retention without approved consent, security and retention policy.
Reviewed recap, targeted document requests, owner / due-date tasks.	Only three nudge types: missing critical fact, understanding gap, workflow event.	Affordability / eligibility conclusions, competitor interpretation and date predictions.	Rejected generative compliance warnings, invented facts and credit decisions.
Compliance event ledger and basic exception queue. Approved scenario display when available.	Natural pauses, cap, dismiss / snooze. Approved scripts and one operational dashboard.	Advanced analytics, audio QA / search and system-wide reconciliation.	Silent material writes, guaranteed dates, "best loan" claims and late dashboard expansion.

Constraint	Fallback that preserves control
Integration unavailable	Reviewed CSV/API work queue, idempotent case ID, original receipt time, daily exception reconciliation
Speech unreliable / too many nudges	Manually added structured notes + reviewed recap. Priority rules, cap, snooze / dismiss and later feedback
High-risk warnings rejected / no audio	Deterministic task routing to named owner. Minimal consent + event log, no raw audio
Dashboard expansion requested	Basic queue and daily export. Capture richer reporting requirements for the next release

Build order 1 Event ledger + Routing flow 2 Capture + review 3 Tasks aid + handoff 4 Approved scenarios